# CareerAI

**AI-Powered Career Intelligence Platform**

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Project: CareerAI — AI-Powered Career Intelligence Platform

## Milestone 1

**SUBMITTED BY:**

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# 1. Introduction

The project "CareerAI — AI-Powered Career Intelligence Platform" is a full-stack web application that functions as an intelligent career coach. It leverages artificial intelligence to analyze resumes, detect skill gaps, predict career paths, and provide real-time AI-powered guidance to help professionals make informed career decisions.

The platform addresses a common challenge faced by job seekers: understanding how their resume performs against applicant tracking systems (ATS), identifying which skills are missing for their target roles, and finding the right career trajectory. CareerAI automates this analysis using NLP-powered resume parsing and large language model inference, delivering actionable insights in seconds.

Users can upload their resume in PDF or DOCX format, receive an ATS compatibility score, view detected skills and skill gaps, explore AI-recommended career paths with salary estimates, browse personalized job and course recommendations, and interact with an AI career coach chatbot that understands their profile context.

# 2. Objective

The objectives of the project are:

- Develop a secure user authentication system.
- Allow users to register and log in safely.
- Enable users to upload resumes in PDF/DOCX format.
- Extract resume text using NLP techniques.
- Store user and resume information securely in the database.
- Build a responsive and user-friendly interface using React.

# 3. Tech Stack

| Layer | Technology | Purpose |
|---|---|---|
| Frontend | React.js + Vite | Fast and responsive single-page application |
| Backend | FastAPI (Python) | REST API development |
| Database | MySQL | Store user and resume data |
| AI Engine | NVIDIA NIM API (Llama 3.1 8B) | AI inference for resume analysis |

# 4. System Architecture

The application follows a three-tier architecture.

### Frontend

- Developed using React.js and Vite.
- Provides Login, Registration, and Resume Upload pages.
- Communicates with the backend using Axios.

### Backend

- Developed using FastAPI.
- Handles authentication using JWT.
- Accepts resume uploads.
- Extracts resume text for AI processing.

### Database

- MySQL stores:
    - User details
    - Authentication data
    - Uploaded resume information

### AI Layer

- NVIDIA NIM API processes extracted resume text for further AI-based analysis.

# 5. Modules Implemented

## 5.1 Landing Page

**Description**

The landing page introduces CareerAI and its purpose. It contains navigation links and provides users with options to sign in or start using the platform.

**Features**

- Responsive navigation bar
- Project introduction
- AI-powered career guidance overview
- Call-to-action buttons (Sign In / Get Started)

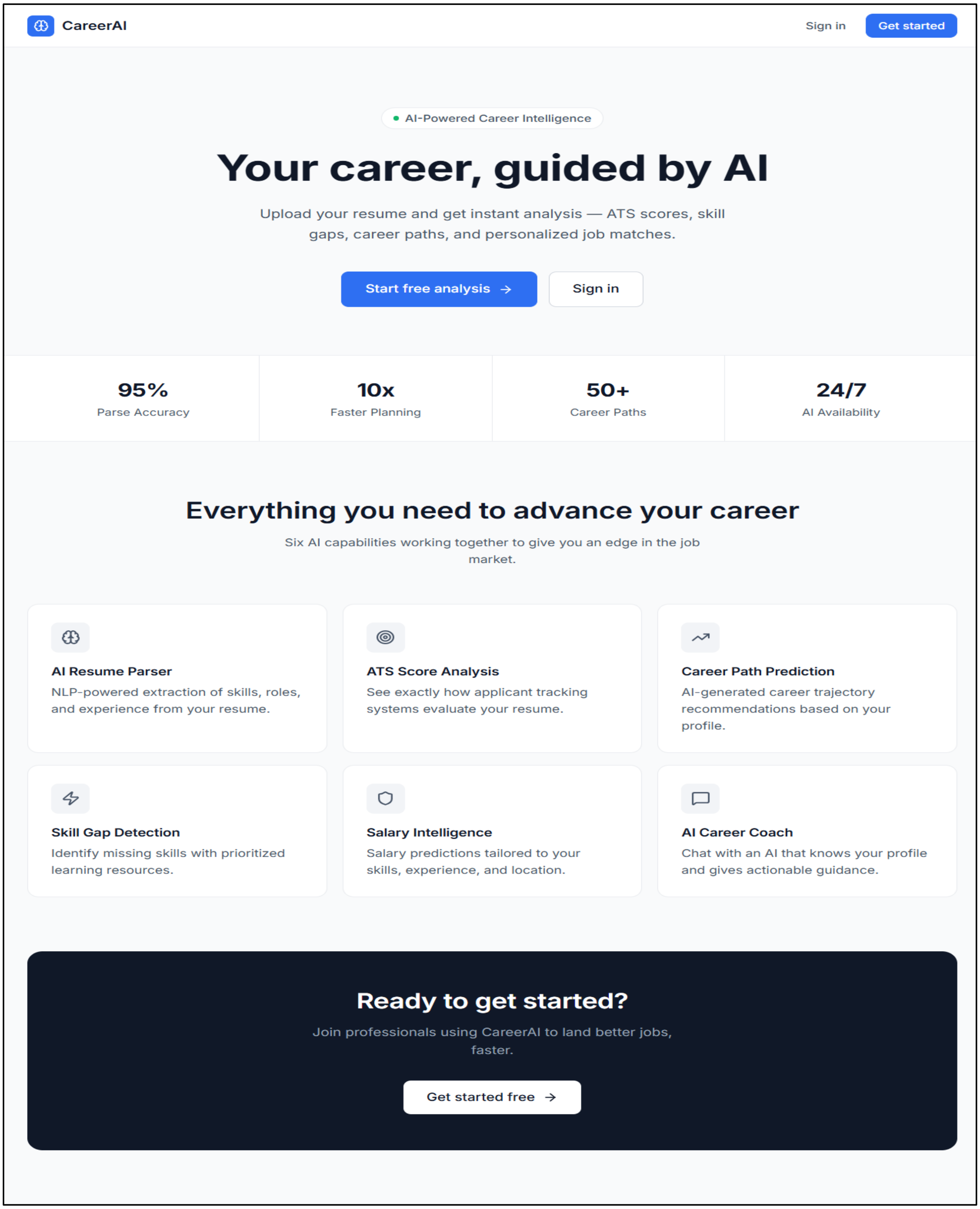



***Figure 1:*** *Landing Page*

## 5.2 Login Page

**Description**

The Login page allows registered users to securely access the CareerAI platform.

**Features**

- Email validation
- Password authentication
- JWT token generation
- Error handling for invalid credentials
- Redirect to Resume Upload page after successful login

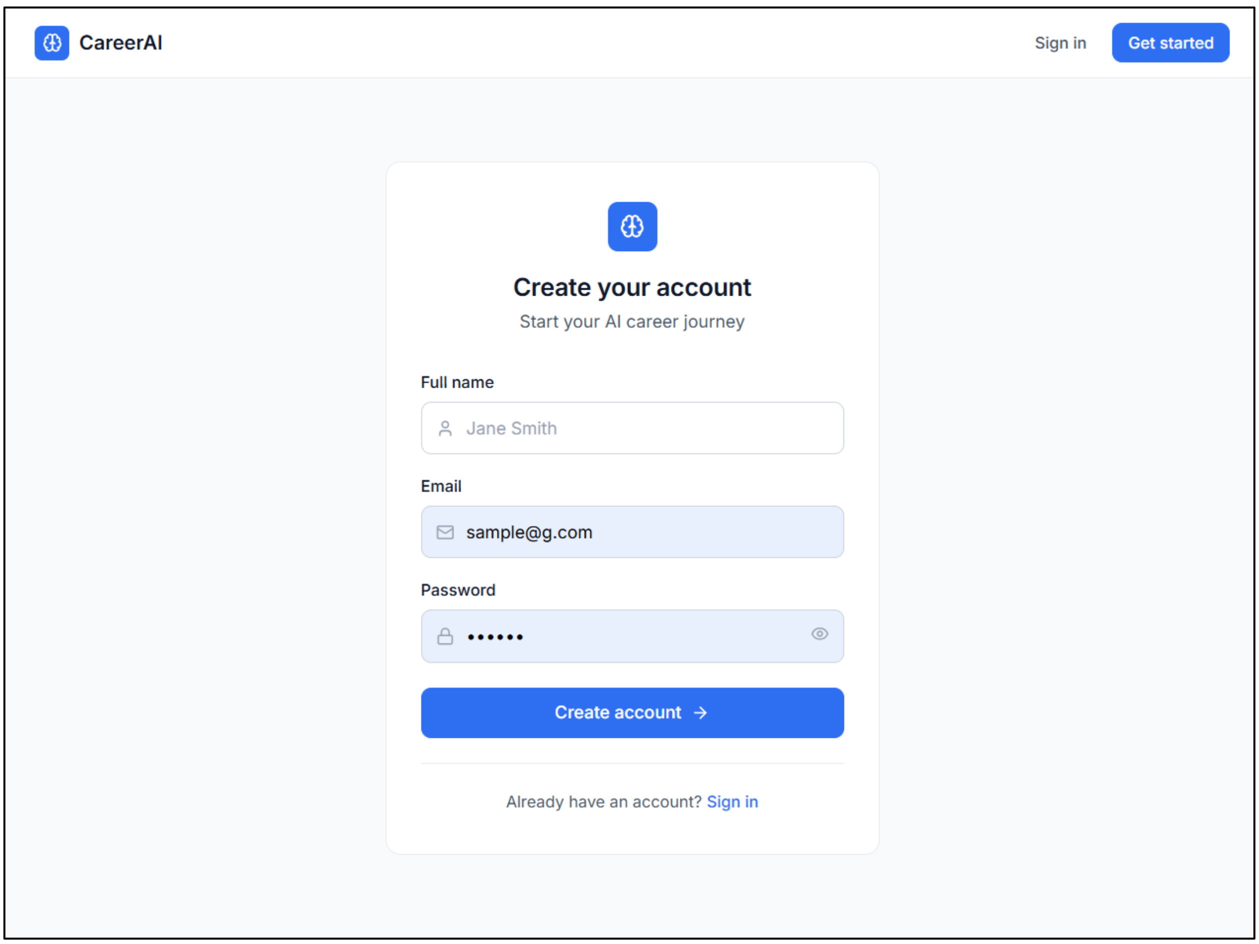



***Figure 2:*** *Login Page*

## 5.3 Resume Upload Page

### Description

The Resume Upload page allows authenticated users to upload resumes for AI processing.

### Features

- Upload PDF/DOCX resumes
- File validation
- Resume stored securely
- Resume sent to backend API
- NLP text extraction initiated
- Success/Error notifications

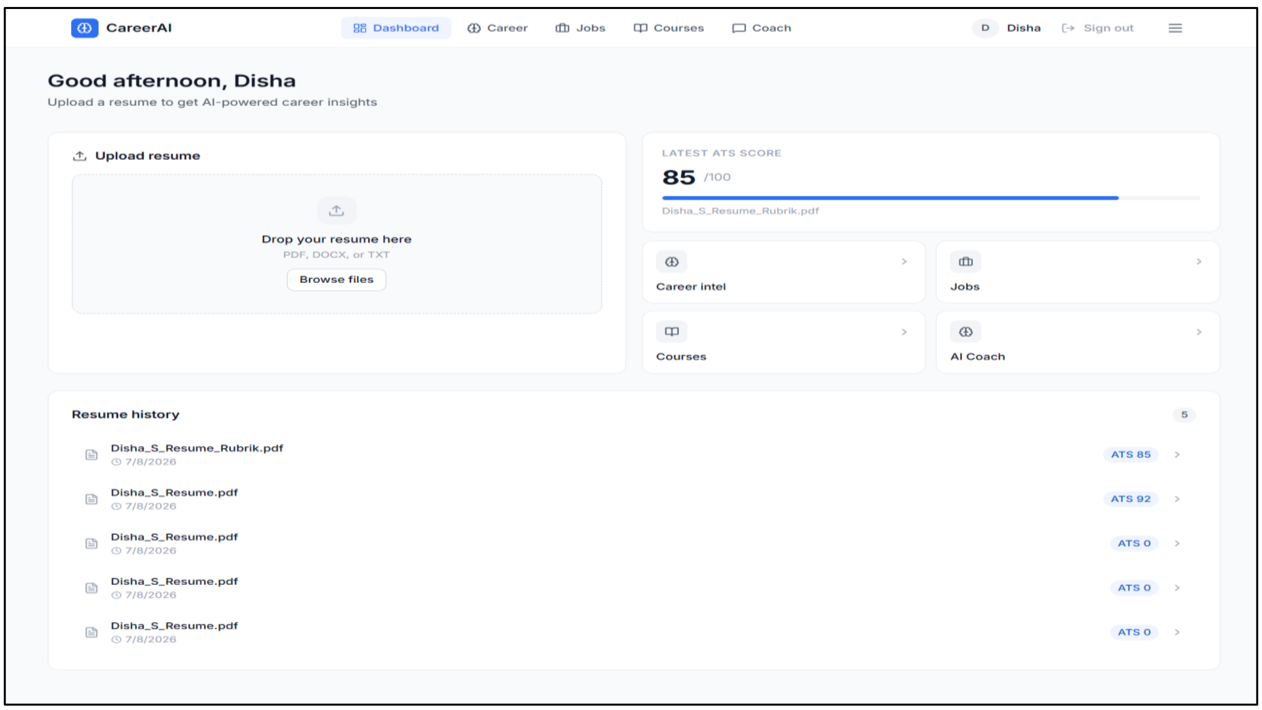



***Figure 3:*** *Resume Upload Page*

## 5.4 User Database

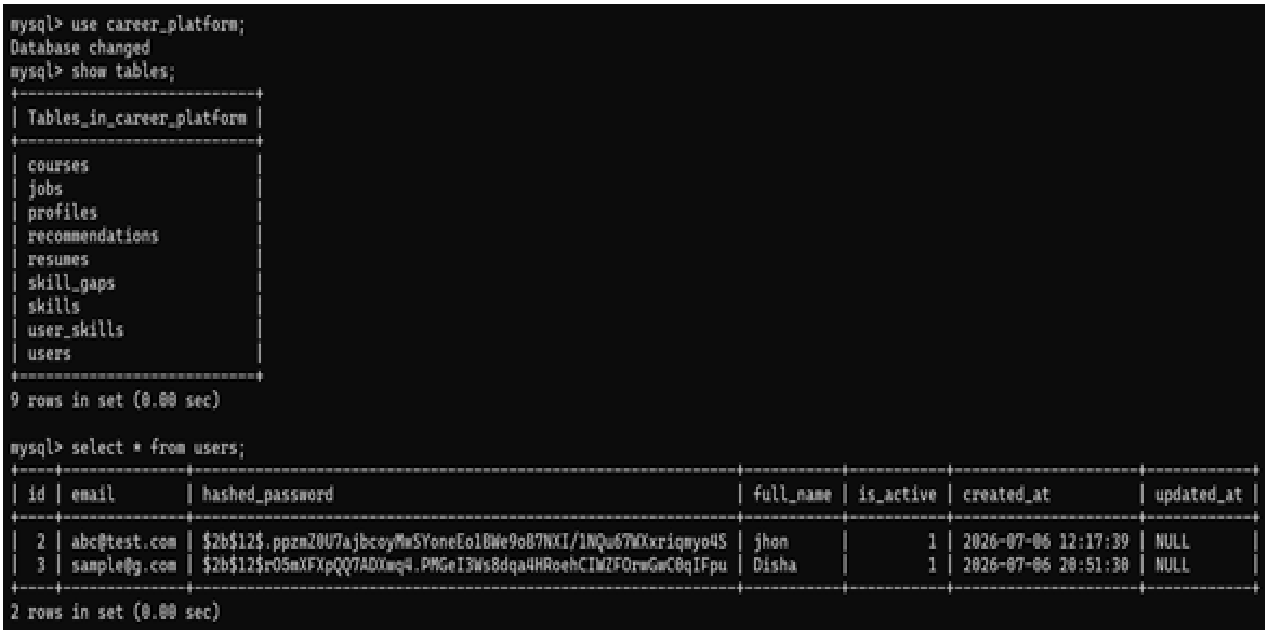
### Description

The User Database stores registered user information and supports secure authentication during login.

### Features

- Stores user details securely

- Passwords stored in hashed format
- Maintains unique user accounts
- Supports JWT-based authentication
- Stores account creation details



```
mysql> use career_platform;
Database changed
mysql> show tables;
+---------------------------+
| Tables_in_career_platform |
+---------------------------+
| courses                   |
| jobs                      |
| profiles                  |
| recommendations           |
| resumes                   |
| skill_gaps                |
| skills                    |
| user_skills               |
| users                     |
+---------------------------+
9 rows in set (0.00 sec)

mysql> select * from users;
+----+---------------+--------------------------------------------------------------+-----------+-----------+---------------------+------------+
| id | email         | hashed_password                                              | full_name | is_active | created_at          | updated_at |
+----+---------------+--------------------------------------------------------------+-----------+-----------+---------------------+------------+
|  2 | abc@test.com  | $2b$12$.ppzmZ0U7ajbcoyMwSYoneEo1BWe9oB7NXI/1NQu67WXxriqmyo4S | jhon      |         1 | 2026-07-06 12:17:39 | NULL       |
|  3 | sample@g.com  | $2b$12$rO5mXFXpQQ7ADXmq4.PMGeI3Ws8dqa4HRoehCIWZFOrwGwC0qIFpu | Disha     |         1 | 2026-07-06 20:51:30 | NULL       |
+----+---------------+--------------------------------------------------------------+-----------+-----------+---------------------+------------+
2 rows in set (0.00 sec)
```

***Figure 4:*** *Database*

## 6. Key Features

### Secure Authentication

- User Registration
- Login
- JWT Authentication

### Resume Upload

- Supports PDF and DOCX files
- Secure file upload
- Resume storage

### Resume Parsing

- Extracts resume text
- Detects skills and experience
- Sends extracted text for AI processing

Responsive User Interface

- Built using React.js
- Mobile-friendly design
- Clean and intuitive interface

## 7. Conclusion

The CareerAI platform successfully implements the initial stages of an AI-powered career guidance system by providing secure user authentication and resume upload functionality. The application establishes a strong foundation for future enhancements such as ATS score analysis, skill gap detection, career path prediction, and AI-based career coaching.